

3D Human Pose Estimation with Spatio-Temporal Criss-cross Attention*

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Abstract

Recent transformer-based solutions have shown great success in 3D human pose estimation. Nevertheless, to calculate the joint-to-joint affinity matrix, the computational cost has a quadratic growth with the increasing number of joints. Such drawback becomes even worse especially for pose estimation in a video sequence, which necessitates spatio-temporal correlation spanning over the entire video. In this paper, we facilitate the issue by decomposing correlation learning into space and time, and present a novel Spatio-Temporal Criss-cross attention (STC) block. Technically, STC first slices its input feature into two partitions evenly along the channel dimension, followed by performing spatial and temporal attention respectively on each partition. STC then models the interactions between joints in an identical frame and joints in an identical trajectory simultaneously by concatenating the outputs from attention layers. On this basis, we devise STCFormer by stacking multiple STC blocks and further integrate a new Structure-enhanced Positional Embedding (SPE) into STCFormer to take the structure of human body into consideration. The embedding function consists of two components: spatio-temporal convolution around neighboring joints to capture local structure, and part-aware embedding to indicate which part each joint belongs to. Extensive experiments are conducted on Human3.6M and MPI-INF-3DHP benchmarks, and superior results are reported when comparing to the state-of-the-art approaches. More remarkably, STCFormer achieves to-date the best published performance: 40.5mm P1 error on the challenging Human3.6M dataset.

1. Introduction

3D human pose estimation has attracted intensive research attention in CV community due to its great poten-

*This work is supported by the National Natural Science Foundation of China under Grants 61932009.

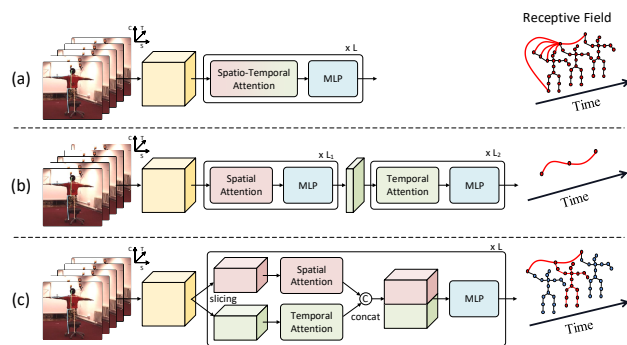


Figure 1. Modeling spatio-temporal correlation for 3D human pose estimation by (a) utilizing spatio-temporal attention on all joints in the entire video, (b) separating the framework into two steps that respectively capture spatial and temporal context, and (c) our Spatio-Temporal Criss-cross attention (STC), i.e., a two-pathway block that models spatial and temporal information in parallel. In the visualization of receptive field, the covered joints of each attention strategy is marked as red nodes.

tial in numerous applications such as human-robot interaction [20, 43], virtual reality [11] and motion prediction [27, 28]. The typical monocular solution is a two-stage pipeline, which first extracts 2D keypoints by 2D human pose detectors (e.g., [7] and [41]), and then lifts 2D coordinates into 3D space [31]. Despite its simplicity, the second stage is an ill-posed problem which lacks the depth prior, and suffers from the ambiguity problem.

To mitigate this issue, several progresses propose to aggregate the temporal cues in a video sequence to promote pose estimation by grid convolutions [15, 26, 35], graph convolutions [4, 47] and multi-layer perceptrons [6, 21]. Recently, Transformer structure has emerged as a dominant architecture in both NLP and CV fields [8, 24, 45, 49], and also demonstrated high capability in modeling spatio-temporal correlation for 3D human pose estimation [13, 22, 23, 25, 48, 52, 54]. Figure 1(a) illustrates a straightforward way to exploit the transformer architecture for directly learning spatio-temporal correlation between all joints in the entire video sequence. However, the computational cost of calcu-

lating the joint-to-joint affinity matrix in the self-attention has a quadratic growth along the increase of number of frames, making such solution unpractical for model training. As a result, most transformer structures employ a two-step alternative, as shown in Figure 1(b), which encodes spatial information for each frame first and then aggregates the feature sequence by temporal transformer. Note that we take spatial transformer as the frame encoder as an example in the figure. This strategy basically mines the correlation across frame-level features but seldom explores the relation between joints across different frames.

In this paper, we propose a novel two-pathway attention mechanism, namely Spatio-Temporal Criss-cross attention (STC), that models spatial and temporal information in parallel, as depicted in Figure 1(c). Concretely, STC first slices the input joint features into two partitions evenly with respect to the channel dimension. On each partition, a Multi-head Self-Attention (MSA) is implemented to encapsulate the context along space or time axis. In between, the space pathway computes the affinity between joints in each frame independently, and the time pathway correlates the identical joint moving across different frames, i.e., the trajectory. Then, STC recombines the learnt contexts from two pathways, and mixes the information across channels by Multi-Layer Perceptrons (MLP). By doing so, the receptive field is like a criss cross of spatial and temporal axes, and the computational cost is $\mathcal{O}(T^2S) + \mathcal{O}(TS^2)$. That is much lower than $\mathcal{O}(T^2S^2)$ of fully spatio-temporal attention, where T and S denote the number of frames and joints, respectively.

By stacking multiple STC blocks, we devise a new architecture — STCFormer for 3D human pose estimation. Furthermore, we delve into the crucial design of positional embedding in STCFormer in the context of pose estimation. The observations that joints in the same body part are either highly relevant (static part) or not relevant but containing moving patterns (dynamic part) motivate us to design a new Structure-enhanced Positional Embedding (SPE). SPE consists of two embedding functions for the static and dynamic part, respectively. A part-aware embedding is to describe the static part by indicating which part each joint belongs to, and a spatio-temporal convolution around neighboring joints aims to capture dynamic structure in local window.

We summarize the main contributions of this work as follows. First, STC is a new type of decomposed spatio-temporal attention for 3D human pose estimation in an economic and effective way. Second, STCFormer is a novel transformer architecture by stacking multiple STC blocks and integrating the structure-enhanced positional embedding. Extensive experiments conducted on Human3.6M and MPI-INF-3DHP datasets demonstrate that STCFormer with much less parameters achieves superior performances than the state-of-the-art techniques.

2. Related Work

Monocular 3D human pose estimation. Monocular 3D human pose estimation is to re-localize human body joints in 3D space from the input single view 2D data, i.e., image or 2D coordinates. The early works [1, 2, 17] develop various graphical or restrictive methods to explore the dependencies of human skeleton and perspective relationships across spaces. With the development of deep learning, several deep neural networks [5, 10, 19, 31, 34, 42, 44, 53] are devised for 3D human pose estimation, and can be categorized into one-stage and two-stage directions. The one-stage approaches directly regress the 3D pose from the input image, and necessitate a large number of image-pose paired data and powerful computing resources [19, 34, 42]. The two-stage methods first exploit off-the-shelf 2D pose detectors [7, 33, 41] to estimate 2D joint coordinates, and then lift the 2D coordinates into 3D space by the fully-connected network [31], grid convolutional network [5], recurrent neural network [10], or graph convolutional network [53]. Although the two-stage methods alleviate the requirement of image-pose pairs, they still heavily suffer from the depth ambiguities problem, which is intrinsically ill-posed due to the lack of depth information.

3D pose estimation from video sequence. To overcome the limitation of depth ambiguities, the advances involve temporal context from neighboring frames to improve 3D coordinates regression. For example, Pavllo *et al.* [35] propose a temporal fully-convolutional network (TCN) to model the local context by convoluting the neighboring frames. Later, Liu *et al.* [26] extend the TCN by introducing an attention mechanism to adaptively identify the significant frames/poses over a sequence. After that, Chen *et al.* [6] decompose the pose estimation into bone length and bone direction prediction. Instead of the aforementioned methods based on temporal aggregation, latter works [4, 16, 46] utilize the spatio-temporal graph convolutional network to model the spatial and temporal correlations across joints simultaneously.

Transformer-based methods. In addition to the traditional convolutional networks, transformer architectures are also exploited to model spatio-temporal correlation [13, 22, 23, 29, 30, 37, 50, 51, 54]. In particular, Zheng *et al.* [54] design a concatenation architecture of several spatial transformer encoders and temporal transformer encoders in PoseFormer. MHFormer [23] proposes to generate multiple hypothesis representations for a pose with the spatial transformer encoder and then model multi-level global correlations with different temporal transformer blocks. StridedFormer [22] and CrossFormer [13] introduce locality by integrating the 1D temporal convolution and 1D spatial convolution, respectively. More recently, the joint-wise inconsistency of motion patterns is highlighted in [48, 52], and encourages to model spatial and temporal information si-

multaneously. PATA [48] groups the joints with similar motion patterns and calculates the intra-part temporal correlation. Similarly, MixSTE [52] uses multiple separated spatial transformer blocks and temporal transformer blocks to model the spatial and temporal correlation iteratively.

Our work also falls into the category of transformer-based method for 3D human pose estimation. The aforementioned transformers mainly model spatial and temporal information respectively in different stages of the networks. In view that the joint motion is a state of coexistence of space and time, such separation may result in insufficient learning of moving patterns. In contrast, our STC block is a two-pathway design that models spatial and temporal dependencies in parallel, which are then mixed through MLP. Moreover, a new positional embedding function is deliberately devised to explore the local structure of human body.

3. Spatio-Temporal Criss-cross Transformer

3.1. Preliminary – Transformer

We begin this section by reviewing the transformer architecture [45] as the basis of our proposal. Transformer is a versatile representation learning architecture, and mainly consists of two components: Multi-head Self-Attention module (MSA) and Feed-Forward Network (FFN). MSA calculates the token-to-token affinity matrix and propagates the information across different tokens. Formally, given N input tokens with C channels, MSA is formulated as

$$MSA(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q} \cdot \mathbf{K}^T}{\sqrt{C}}\right) \cdot \mathbf{V}, \quad (1)$$

where $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{N \times C}$ denote the queries, keys and values obtained by linearly mapping the input tokens. Note that we omit the multi-head separation here for simplicity. FFN contains a Multi-Layer Perceptrons (MLP), i.e., a non-linear mapping with two linear layer plus a GELU [14] activation in between. The output of MLP is computed by

$$MLP(\mathbf{H}) = \text{GELU}(\mathbf{H} \cdot \mathbf{W}_1) \cdot \mathbf{W}_2, \quad (2)$$

where $\mathbf{H} \in \mathbb{R}^{N \times C}$ is the input tokens of MLP, $\mathbf{W}_1 \in \mathbb{R}^{C \times \hat{C}}$ and $\mathbf{W}_2 \in \mathbb{R}^{\hat{C} \times C}$ are the projection matrices. With these, each transformer block is constructed by utilizing MSA and MLP in order with shortcut connection:

$$\begin{aligned} \mathbf{Q}, \mathbf{K}, \mathbf{V} &= FC(LN(\mathbf{X})), \\ \mathbf{Y} &= MSA(\mathbf{Q}, \mathbf{K}, \mathbf{V}) + \mathbf{X}, \\ \mathbf{Z} &= MLP(LN(\mathbf{Y})) + \mathbf{Y}, \end{aligned} \quad (3)$$

where FC is linear projection of the input tokens $\mathbf{X}$, and LN denotes Layer Norm [3]. The output $\mathbf{Z}$ serves as the input to the next block until the last one.

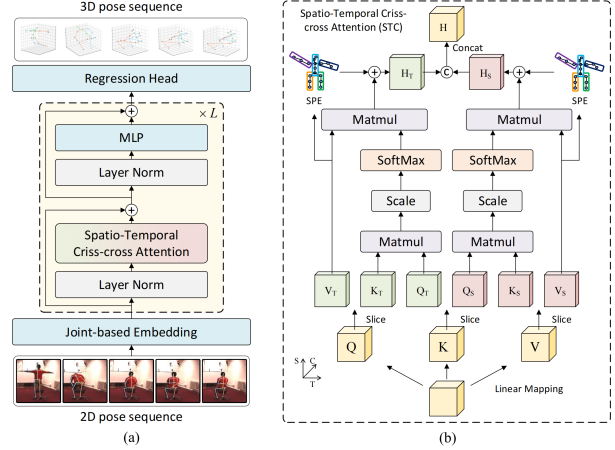


Figure 2. An overview of our proposed Spatio-Temporal Criss-cross Transformer (STCFormer). (a) It mainly consists of L sequential STC blocks. Each block aggregates the context across tokens by spatio-temporal criss-cross attention, and non-linearly maps each token by Multi-Layer Perceptrons (MLP). (b) The architecture of our STC block and the Structure-enhanced Positional Embedding (SPE).

3.2. Overall Architecture

Figure 2 depicts an overview of the proposed STCFormer, which mainly includes three stages: a joint-based embedding, stacked STC blocks and a regression head. The joint-based embedding projects the input 2D coordinates of each joint into feature space. STC blocks aggregate the spatio-temporal context, and update the representation of each joint. Based on the learnt features, the 3D coordinates are estimated by a regression head.

Joint-based embedding. Given a 2D pose sequence as $\mathbf{P}_{2D} \in \mathbb{R}^{T \times N \times 2}$, where T and N denote the number of frames and the number of body joints in each frame, respectively, we first project $\mathbf{P}_{2D}$ to high-dimensional embeddings by a joint-based embedding layer. This layer applies an FC layer to each 2D coordinate independently followed by a GELU activation. As such, the joint-based embedding layer produces the features with the shape of $T \times N \times C$. Note that in the previous transformer [22], the embedding layer projects all joint coordinates in each frame into a single vector, reducing the computational cost of the subsequent transformer blocks while losing the spatial discrimination. Ours is different in that the spatial dimension N is maintained, and the computational cost is also reduced by spatio-temporal criss-cross attention.

STC blocks. The STC block originates from the transformer block in Eq.(3), and replaces the original MSA layer with spatio-temporal criss-cross attention. In addition, a new positional embedding function, i.e., Structure-enhanced Positional Embedding (SPE), is integrated into the STC block for better descriptive capability of local structures. Section 3.3 and Section 3.4 will elaborate STC and SPE, respectively.

Regression head. A liner regression head is finally established upon the STC blocks to estimate the 3D pose coordinates $\hat{\mathbf{P}}_{3D} \in \mathbb{R}^{T \times N \times 3}$. The whole architecture is optimized by minimizing the Mean Squared Error (MSE) between $\hat{\mathbf{P}}_{3D}$ and the ground-truth 3D coordinates $\mathbf{P}_{3D}$ as

$$\mathcal{L} = \left\| \hat{\mathbf{P}}_{3D} - \mathbf{P}_{3D} \right\|^2. \quad (4)$$

3.3. Spatio-Temporal Criss-cross Attention

STC aims to model the spatio-temporal dependencies between joints in an efficient way to avoid the quadratic computation cost of fully spatio-temporal attention. Inspired by the group contextualization strategy [12] which separates the channels into several paralleled groups and applies different feature contextualization operations to them respectively, we propose to capture the spatial and temporal context on different channels in parallel. Different from the axial convolution in [12, 36, 38], we exploit axis-specific multi-head self-attention in STC for spatial or temporal context, which is more powerful for correlation learning.

Concretely, the input embedding $\mathbf{X} \in \mathbb{R}^{T \times N \times C}$ are firstly mapped to queries $\mathbf{Q} \in \mathbb{R}^{T \times N \times C}$, keys $\mathbf{K} \in \mathbb{R}^{T \times N \times C}$, and values $\mathbf{V} \in \mathbb{R}^{T \times N \times C}$, which are then evenly divided into two groups along the channel dimension. For notation clarity, we denote the divided feature matrix as time group $\{\mathbf{Q}_T, \mathbf{K}_T, \mathbf{V}_T\}$ and space group $\{\mathbf{Q}_S, \mathbf{K}_S, \mathbf{V}_S\}$. Next, the temporal and spatial correlations are calculated in two separate self-attention modules.

Temporal correlation represents the relation between the joints in an identical trajectory moving across different frames. To achieve this, we implement an axis-specific MSA, named MSA_T , which computes the attention affinities in Eq.(1) between joints across the temporal dimension. Hence, the output of temporal attention is measured as

$$\mathbf{H}_T = MSA_T(\mathbf{Q}_T, \mathbf{K}_T, \mathbf{V}_T). \quad (5)$$

Spatial correlation is the connection between joints in an identical frame. These joints indicate different body parts in one frame, which are intrinsically relevant due to the prior of body skeleton. Similar to temporal attention, we devise MSA_S as an axis-specific MSA component on spatial dimension. Therefore, the output of spatial attention is formulated as

$$\mathbf{H}_S = MSA_S(\mathbf{Q}_S, \mathbf{K}_S, \mathbf{V}_S). \quad (6)$$

The above two correlation modules process in parallel and follow the self-attention regime for feature contextualization. They compute the token-to-token affinities by contextualizing from a specific axial perspective, and complement to each other. Thus, we concatenate the outputs from both attention layers along the channel dimension:

$$\mathbf{H} = \text{cat}(\mathbf{H}_T, \mathbf{H}_S), \quad (7)$$

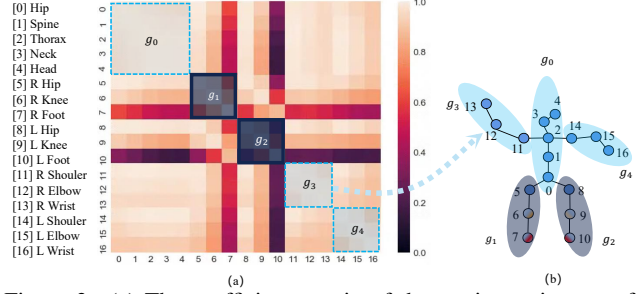


Figure 3. (a) The coefficient matrix of the motion trajectory of different joints. (b) The body joints are divided into five parts, denoted as g_* . The part with high/low relevance is colored as light/dark blue, respectively. The motion data is generated by actor S6 performing greeting action in the training set of Human3.6M.

where cat performs the concatenation. The resultant receptive field of STC is like a criss cross of spatial and temporal axes, and stacking multiple STC blocks is able to approximate the fully spatio-temporal attention.

3.4. Structure-enhanced Positional Embedding

One of the crucial factor in transformer is positional embedding, which indicates the position of each token absolutely or relatively. For the positional embedding function in STCFormer, we delve into the inherent property of joints, i.e., the local structure, and propose Structure-enhanced Positional Embedding (SPE). Figure 3 depicts the motivation of SPE. Here, we group the body joints into five parts according to the dynamic chain structure of human body:

$$\begin{aligned} g_0 &= \{\text{hip, spine, thorax, neck, head}\} \\ g_1 &= \{\text{right_hip, right_knee, right_feet}\} \\ g_2 &= \{\text{left_hip, left_knee, left_feet}\} \\ g_3 &= \{\text{right_shoulder, right_elbow, right_wrist}\} \\ g_4 &= \{\text{left_shoulder, left_elbow, left_wrist}\} \end{aligned} \quad (8)$$

The trajectories of joints in the **static part** (g_0 , g_3 and g_4 in the figure) are highly relevant. We devise a part-wise positional embedding to indicate which part each joint belongs to. The joints in the same part are attached with the same embedding vector. In particular, a learnable dictionary is constructed to assign embedding vector to different joints according to their group index. Given the group index $\mathbf{g} \in [0, 1, 2, 3, 4]^{T \times N}$ of joints, the learnable dictionary $\mathbf{D} \in \mathbb{R}^{5 \times \frac{C}{2}}$ convert the indexes to embedding vectors as

$$\mathbf{SPE}_1 = \mathbf{D}(\mathbf{g}). \quad (9)$$

Nevertheless, in the **dynamic part**, i.e., part with relative movements (g_1 , g_2 in the figure), the trajectories of joints are not relevant. Simply assigning the same embedding vector to these joints ignores the motion patterns in the dynamic part. Hence, we propose to exploit a spatio-temporal convolution around the neighboring joints to capture the local structure. Formally, given the values $\mathbf{V} \in \mathbb{R}^{T \times N \times \frac{C}{2}}$ in

Algorithm 1 Pseudo-code of STC with SPE (PyTorch-like)

```
# x: input tensor of shape (B, T, N, C)
# p: part index (B, T, N) in [0, 4]
# MSA: axis-specific multi-head self-attention
self.linear = nn.Linear(C, 3C)
self.embed1 = nn.Embedding(5, C//2)
# the channel-last convolution
self.embed2 = nn.Conv2d(C//2, C//2, k=3, g=C//2)

def STC(x, p):
    Q, K, V = self.linear(x).chunk(3, dim=3)
    Q_t, Q_s = Q.chunk(2, dim=3)
    K_t, K_s = K.chunk(2, dim=3)
    V_t, V_s = V.chunk(2, dim=3)

    H_t = MSA(Q_t, K_t, V_t, dim=1)
    H_s = MSA(Q_s, K_s, V_s, dim=2)

    H_t += self.embed1(p) + self.embed2(V_t)
    H_s += self.embed1(p) + self.embed2(V_s)

    H = torch.cat(H_t, H_s, dim=3)
    return H
```

STC block, we treat $\mathbf{V}$ as 2D (i.e., space and time) feature map, and utilize 2D convolution on the neighboring joints:

$$\mathbf{SPE}_2(\mathbf{V}) = \text{conv2d}(\mathbf{V}), \quad (10)$$

where conv2d is a 3×3 convolution operation. Although the two SPE functions are designed respectively for static part and dynamic part, we utilize the two functions concurrently on all joints leaving out the requirement of static/dynamic judgment. The duet of two SPE functions is able to deal with the parts with various moving patterns.

By injecting the proposed SPE function into STC, the equation of STC is reformulated as

$$\begin{aligned} \mathbf{H}_T &= \text{MSA}_T(\mathbf{Q}_T, \mathbf{K}_T, \mathbf{V}_T) + \mathbf{SPE}_1 + \mathbf{SPE}_2(\mathbf{V}_T), \\ \mathbf{H}_S &= \text{MSA}_S(\mathbf{Q}_S, \mathbf{K}_S, \mathbf{V}_S) + \mathbf{SPE}_1 + \mathbf{SPE}_2(\mathbf{V}_S), \\ \mathbf{H} &= \text{cat}(\mathbf{H}_T, \mathbf{H}_S). \end{aligned} \quad (11)$$

Implementation. The proposed STC plus SPE in Eq.(11) can be readily implemented with a few lines of codes in Python. We detail an example of the codes in Algorithm 1 based on PyTorch platform. Here, we execute the pre-defined MSA and MLP function in the standard transformer. The SPE is implemented by constructing the default Embedding layer and Conv2d layer.

4. Experiments

We comprehensively evaluate the proposed STCFormer architecture on two large-scale datasets, i.e., Human3.6M [18] and MPI-INF-3DHP [32].

4.1. Datasets and Evaluation Metrics

Human3.6M is currently the most popular benchmark for indoor 3D human pose estimation, which contains 11

subjects performing 15 typical actions, leading to 3.6 million video frames in total. Following the standard protocol, we use subjects 1, 5, 6, 7, and 8 for training, and subjects 9 and 11 for evaluation. The Mean Per Joint Position Error (MPJPE) is used to measure the error under two protocols: Protocol 1 (referred to as P1) computes MPJPE between the estimated pose and the ground truth after aligning their root joints (hip); Protocol 2 (referred to as P2) calculates Procrustes-MPJPE, where the ground truth and the pose prediction are further aligned through a rigid transformation. We also compute the MPJPE distribution of pose to evaluate the overall precision of the reconstructed skeletons. **MPI-INF-3DHP** is a recently proposed large-scale dataset, which consists of three scenes, i.e., green screen, non-green screen, and outdoor. By using 14 cameras, the dataset records 8 actors performing 8 activities for the training set and 7 activities for evaluation. Following the previous works [6, 39, 54], we adopt the MPJPE (P1), percentage of correct keypoints (PCK) with 150mm, and area under the curve (AUC) results as the evaluation metrics.

4.2. Implementation Details

Our model is implemented with PyTorch toolkit and runs on a server with one GTX 2080Ti GPU. In the experiments, two kinds of input 2D pose sequences are utilized including the pre-estimated 2D pose by the pre-trained CPN [7] and the real 2D pose (ground truth). For model training, we set each mini-batch as 128 sequences. The network parameters are optimized for 20 epochs by Adam optimizer with basic learning rate of 0.001 and decayed by 0.96 after each epoch. We consider the repeat time L of modules, the hidden embedding channel C , and the number of head H in attention block as free parameters that we tailor to the scale of network. The performances of the standard version STCFormer with $\{L=6, C=256, H=8\}$ and the large version STCFormer-L with $\{L=6, C=512, H=8\}$ are both reported.

4.3. Performance Comparison on Human3.6M

We compare with several state-of-the-art techniques on Human3.6M dataset. Table 1 summarizes the performance comparisons in terms of P1 and P2 errors taking the pre-estimated 2D poses (CPN) as input, and the number of sampled frames T per video is also given for each method. In general, the longer input sequence leads to the lower regression error. Overall, STCFormer-L with $T=243$ input frames achieves the new state-of-the-art performances with P1 error of 40.5mm and P2 error of 31.8mm. Benefiting from the proposed STC attention module, STCFormer-L outperforms StridedFormer [22], PATA [48] and MixSTE [52] with $T=243$ frames, which are also based on transformer architecture, by the P1 error drop of 3.2mm, 2.6mm and 0.4mm, respectively. Comparing to the best competitor

Table 1. Performance comparisons in terms of P1 error (mm) and P2 error (mm) with the state-of-the-art methods on Human3.6M dataset. The 2D pose input is estimated by CPN [7]. The best result and runner-up result in each column are marked in red and blue, respectively. “*” denotes the post-processing module proposed in [4]. T is the number of sampled frames from each video.

P1	Publication	Dir.	Dis.	Eat.	Gre.	Phone	Photo	Pose	Purch.	Sit.	SitD.	Smoke	Wait	WalkD.	Walk	WalkT.	Avg.
Liu <i>et al.</i> [26] ($T=243$)	CVPR’20	41.8	44.8	41.1	44.9	47.4	54.1	43.4	42.2	56.2	63.6	45.3	43.5	45.3	31.3	32.2	45.1
UGCN [46] ($T=96$) *	ECCV’20	40.2	42.5	42.6	41.1	46.7	56.7	41.4	42.3	56.2	60.4	46.3	42.2	46.2	31.7	31.0	44.5
PoseFormer [54] ($T=81$)	ICCV’21	41.5	44.8	39.8	42.5	46.5	51.6	42.1	42.0	53.3	60.7	45.5	43.3	46.1	31.8	32.2	44.3
Shan <i>et al.</i> [40] ($T=243$)	ACM MM’21	40.8	44.5	41.4	42.7	46.3	55.6	41.8	41.9	53.7	60.8	45.0	41.5	44.8	30.8	31.9	44.3
Anatomy3D [6] ($T=243$)	TCVST’21	41.4	43.5	40.1	42.9	46.6	51.9	41.7	42.3	53.9	60.2	45.4	41.7	46.0	31.5	32.7	44.1
Einfalt <i>et al.</i> [9] ($T=351$) *	arXiv’22	39.6	43.8	40.2	42.4	46.5	53.9	42.3	42.5	55.7	62.3	45.1	43.0	44.7	30.1	30.8	44.2
StridedFormer [22] ($T=243$) *	TMM’22	40.3	43.3	40.2	42.3	45.6	52.3	41.8	40.5	55.9	60.6	44.2	43.0	44.2	30.0	30.2	43.7
CrossFormer [13] ($T=81$)	arXiv’22	40.7	44.1	40.8	41.5	45.8	52.8	41.2	40.8	55.3	61.9	44.9	41.8	44.6	29.2	31.1	43.7
PATA [48] ($T=243$)	TIP’22	39.9	42.7	40.3	42.3	45.0	52.8	40.4	39.3	56.9	61.2	44.1	41.3	42.8	28.4	29.3	43.1
MHFormer [23] ($T=351$)	CVPR’22	39.2	43.1	40.1	40.9	44.9	51.2	40.6	41.3	53.5	60.3	43.7	41.1	43.8	29.8	30.6	43.0
P-STMO [39] ($T=243$)	ECCV’22	38.9	42.7	40.4	41.1	45.6	49.7	40.9	39.9	55.5	59.4	44.9	42.2	42.7	29.4	29.4	42.8
MixSTE [52] ($T=81$)	CVPR’22	39.8	43.0	38.6	40.1	43.4	50.6	40.6	41.4	52.2	56.7	43.8	40.8	43.9	29.4	30.3	42.4
MixSTE [52] ($T=243$)	CVPR’22	37.6	40.9	37.3	39.7	42.3	49.9	40.1	39.8	51.7	55.0	42.1	39.8	41.0	27.9	27.9	40.9
STCFormer ($T=81$)		40.6	43.0	38.3	40.2	43.5	52.6	40.3	40.1	51.8	57.7	42.8	39.8	42.3	28.0	29.5	42.0
STCFormer ($T=243$)		39.6	41.6	37.4	38.8	43.1	51.1	39.1	39.7	51.4	57.4	41.8	38.5	40.7	27.1	28.6	41.0
STCFormer-L ($T=243$)		38.4	41.2	36.8	38.0	42.7	50.5	38.7	38.2	52.5	56.8	41.8	38.4	40.2	26.2	27.7	40.5
P2	Publication	Dir.	Dis.	Eat.	Gre.	Phone	Photo	Pose	Purch.	Sit.	SitD.	Smoke	Wait	WalkD.	Walk	WalkT.	Avg.
Liu <i>et al.</i> [26] ($T=243$)	CVPR’20	32.3	35.2	33.3	35.8	35.9	41.5	33.2	32.7	44.6	50.9	37.0	32.4	37.0	25.2	27.2	35.6
UGCN [46] ($T=96$) *	ECCV’20	31.8	34.3	35.4	33.5	35.4	41.7	31.1	31.6	44.4	49.0	36.4	32.2	35.0	24.9	23.0	34.5
PoseFormer [54] ($T=81$)	ICCV’21	34.1	36.1	34.4	37.2	36.4	42.2	34.4	33.6	45.0	52.5	37.4	33.8	37.8	25.6	27.3	36.5
Shan <i>et al.</i> [40] ($T=243$)	ACM MM’21	32.5	36.2	33.2	35.3	35.6	42.1	32.6	31.9	42.6	47.9	36.6	32.1	34.8	24.2	25.8	35.0
Anatomy3D [6] ($T=243$)	TCSVT’21	32.6	35.1	32.8	35.4	36.3	40.4	32.4	32.3	42.7	49.0	36.8	32.4	36.0	24.9	26.5	35.0
Einfalt <i>et al.</i> [9] ($T=351$) *	arXiv’22	32.7	36.1	33.4	36.0	36.1	42.0	33.3	33.1	45.4	50.7	37.0	34.1	35.9	24.4	25.4	35.7
StridedFormer [22] ($T=243$) *	TMM’22	32.7	35.5	32.5	35.4	35.9	41.6	33.0	31.9	45.1	50.1	36.3	33.5	35.1	23.9	25.0	35.2
MHFormer [23] ($T=351$)	CVPR’22	31.5	34.9	32.8	33.6	35.3	39.6	32.0	32.2	43.5	48.7	36.4	32.6	34.3	23.9	25.1	34.4
P-STMO [39] ($T=243$)	ECCV’22	31.3	35.2	32.9	33.9	35.4	39.3	32.5	31.5	44.6	48.2	36.3	32.9	34.4	23.8	23.9	34.4
CrossFormer [13] ($T=81$)	arXiv’22	31.4	34.6	32.6	33.7	34.3	39.7	31.6	31.0	44.3	49.3	35.9	31.3	34.4	23.4	25.5	34.3
PATA [48] ($T=243$)	TIP’22	31.2	34.1	31.9	33.8	33.9	39.5	31.6	30.0	45.4	48.1	35.0	31.1	33.5	22.4	23.6	33.7
MixSTE [52] ($T=81$)	CVPR’22	32.0	34.2	31.7	33.7	34.4	39.2	32.0	31.8	42.9	46.9	35.5	32.0	34.4	23.6	25.2	33.9
MixSTE [52] ($T=243$)	CVPR’22	30.8	33.1	30.3	31.8	33.1	39.1	31.1	30.5	42.5	44.5	34.0	30.8	32.7	22.1	22.9	32.6
STCFormer ($T=81$)		30.4	33.8	31.1	31.7	33.5	39.5	30.8	30.0	41.8	45.8	34.3	30.1	32.8	21.9	23.4	32.7
STCFormer ($T=243$)		29.5	33.2	30.6	31.0	33.0	38.0	30.4	29.4	41.8	45.2	33.6	29.5	31.6	21.3	22.6	32.0
STCFormer-L ($T=243$)		29.3	33.0	30.7	30.6	32.7	38.2	29.7	28.8	42.2	45.0	33.3	29.4	31.5	20.9	22.3	31.8

Table 2. Performance comparisons in terms of P1 error (mm) with the state-of-the-art methods on Human3.6M dataset. The models take the ground-truth 2D pose as input. The best result and runner-up result in each column are marked in red and blue, respectively. “*” denotes the post-processing module proposed in [4]. T is the number of sampled frames from each video.

P1	Publication	Dir.	Dis.	Eat.	Gre.	Phone	Photo	Pose	Purch.	Sit.	SitD.	Smoke	Wait	WalkD.	Walk	WalkT.	Avg.
Liu <i>et al.</i> [26] ($T=243$)	CVPR’20	34.5	37.1	33.6	34.2	32.9	37.1	39.6	35.8	40.7	41.4	33.0	33.8	33.0	26.6	26.9	34.7
PoseFormer [54] ($T=81$)	ICCV’21	30.0	33.6	29.9	31.0	30.2	33.3	34.8	31.4	37.8	38.6	31.7	31.5	29.0	23.3	23.1	31.3
Shan <i>et al.</i> [40] ($T=243$)	ACM MM’21	29.5	30.8	28.8	29.1	30.7	35.2	31.7	27.8	34.5	36.0	30.3	29.4	28.9	24.1	24.7	30.1
MHFormer [23] ($T=351$)	CVPR’22	27.7	32.1	29.1	28.9	30.0	33.9	33.0	31.2	37.0	39.3	30.0	31.0	29.4	22.2	23.0	30.5
P-STMO [39] ($T=243$)	ECCV’22	28.5	30.1	28.6	27.9	29.8	33.2	31.3	27.8	36.0	37.4	29.7	29.5	28.1	21.0	21.0	29.3
StridedFormer [22] ($T=243$) *	TMM’22	27.1	29.4	26.5	27.1	28.6	33.0	30.7	26.8	38.2	34.7	29.1	29.8	26.8	19.1	19.8	28.5
CrossFormer [13] ($T=81$)	arXiv’22	26.0	30.0	26.8	26.2	28.0	31.0	30.4	29.6	35.4	37.1	28.4	27.3	26.7	20.5	19.9	28.3
PATA [48] ($T=243$)	TIP’22	25.8	25.2	23.3	23.5	24.0	27.4	27.9	24.4	29.3	30.1	24.9	24.1	23.3	18.6	19.7	24.7
MixSTE [52] ($T=81$)	CVPR’22	25.6	27.8	24.5	25.7	24.9	29.9	28.6	27.4	29.9	29.0	26.1	25.0	25.2	18.7	19.9	25.9
MixSTE [52] ($T=243$)	CVPR’22	21.6	22.0	20.4	21.0	20.8	24.3	24.7	21.9	26.9	24.9	21.2	21.5	20.8	14.7	15.7	21.6
STCFormer ($T=81$)		26.2	26.5	23.4	24.6	25.0	28.6	28.3	24.6	30.9	33.7	25.7	25.3	24.6	18.6	19.7	25.7
STCFormer ($T=81$) *		25.9	25.9	22.7	24.0	24.6	27.5	27.6	23.1	30.1	31.5	25.1	24.7	23.8	18.4	19.6	25.0
STCFormer ($T=243$)		21.4	22.6	21.0	21.3	23.8	26.0	24.2	20.0	28.9	28.0	22.3	21.4	20.1	14.2	15.0	22.0
STCFormer ($T=243$) *		20.8	21.8	20.0	20.6	23.4	25.0	23.6	19.3	27.8	26.1	21.6	20.6	19.5	14.3	15.1	21.3

MixSTE [52], our STCFormer consistently obtains better precision across different numbers of input frames, and only demands around half of the parameters (18.9M v.s. 33.6M). The results verify the advantages of STC attention as an economic and effective way to decompose the full spatio-temporal attention. More importantly, the series of STCFormer reaches to-date the best reported performances in 10 out of 15 categories.

Table 2 further details the comparisons between STCFormer and the state-of-the-art models with the ground-truth 2D pose as input. This setting excludes the noise from 2D pose estimation, and measures the upper bound of 2D-to-3D lifting models. Accordingly, the P1 errors are obvi-

ously decreased across different methods by replacing the CPN-estimated 2D pose with the ground-truth 2D pose, but the performance trends are still similar. STCFormer with post-processing attains the best P1 error of 21.3mm, which is 0.3mm lower than the best competitor MixSTE, validating the impact of STCFormer with different types of input.

In addition to the mean error, we also compare the error distribution of STCFormer and baseline methods in Figure 4. In this experiment, the methods take the estimated 2D poses by CPN of 27 frames as input. Compared to the recent transformer-based approaches including StridedFormer [22], P-STMO [39], and MHFormer [23], our STCFormer leads to the highest number of samples with error

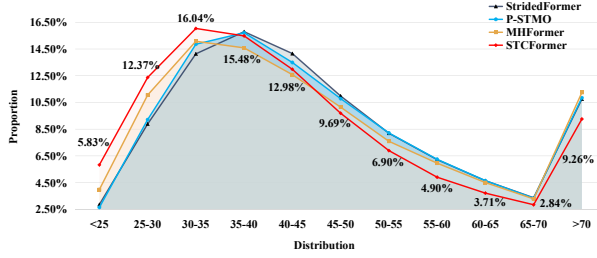


Figure 4. Error distribution of the estimated 3D poses on Human3.6M. The horizontal axis represents the error interval, and the vertical axis is the proportion of poses with error in the interval.

less than 35mm, and the lowest number of those with error larger than 45mm. This again confirm the advances of STCFormer for not only obtaining the lowest average error but also better distribution across different ranges of error.

4.4. Performance Comparison on MPI-INF-3DHP

To verify the generalization of 3D pose estimation models, we then test the performance on MPI-INF-3DHP dataset, which contains more complex backgrounds. Following previous works [23, 39, 52], the ground-truth 2D poses are taken as input. In view of the shorter video sequence, we set the number of input frames as 9, 27 or 81. Table 3 lists the performance comparisons. Similar to the observations on Human3.6M, our STCFormer with $T=81$ reaches the to-date best reported performance with PCK of 98.7%, AUC of 83.9% and P1 error of 23.1mm, outperforming the current state-of-the-art models with a large margin of 0.8% in PCK, 8.1% in AUC and 9.1mm in P1 error. In particular, STCFormer shows better generalization ability and surpasses MixSTE [52] by a much larger P1 error drop (31.8mm) against 0.3mm on Human3.6M. This highlights the efficacy of our method on the more complicated data.

4.5. Ablation Study

For a more in-depth analysis of our STCFormer, we further conduct a series of ablation studies on Human3.6M dataset using the CPN-estimated 2D poses as input.

The first group of experiments is to verify how well our STCFormer works with different number of input frames. Table 4 shows the detailed comparisons in terms of P1 error. A general performance tendency is observed that increasing T leads to monotonic performance improvement. Among the competitive methods, our STCFormer constantly exhibits the best results across 27-frame, 81-frame and 243-frame settings. The leading performances demonstrate the ability of STCFormer to deal with different length of video sequence. More remarkably, STCFormer-L has 43.7% fewer parameters and spends 43.6% fewer FLOPs than the runner-up MixSTE.

The second ablation study assesses the performance impact of different design components. In this experiment, the models take the estimated 2D poses by CPN of 27 frames

Table 3. Performance comparisons in terms of PCK, AUC and P1 with the state-of-the-art methods on MPI-INF-3DHP dataset. Here, the higher PCK, the higher AUC and the lower P1 indicate the better regressions. The best result in each column is marked in red. T is the number of sampled frames from each video.

Method	Publication	PCK $\uparrow$	AUC $\uparrow$	P1(mm) $\downarrow$
UGCN [46] ($T=96$)	ECCV'20	86.9	62.1	68.1
Anatomy3D [6] ($T=81$)	TCSVT'21	87.8	53.8	79.1
PoseFormer [54] ($T=9$)	ICCV'21	88.6	56.4	77.1
Hu <i>et al.</i> [16] ($T=96$)	ACM MM'21	97.9	69.5	42.5
CrossFormer [13] ($T=9$)	arXiv'22	89.1	57.5	76.3
PATA [48] ($T=243$)	TIP'22	90.3	57.8	69.4
MHFormer [23] ($T=9$)	CVPR'22	93.8	63.3	58.0
MixSTE [52] ($T=27$)	CVPR'22	94.4	66.5	54.9
Einfalt <i>et al.</i> [9] ($T=81$)	arXiv'22	95.4	67.6	46.9
P-STMO [39] ($T=81$)	ECCV'22	97.9	75.8	32.2
STCFormer ($T=9$)		98.2	81.5	28.2
STCFormer ($T=27$)		98.4	83.4	24.2
STCFormer ($T=81$)		98.7	83.9	23.1

Table 4. The P1 error comparisons with different number of sampled frame (T) on Human3.6M dataset. The best result in each column is marked in red.

Method	Frames T	Parameters	FLOPs (M)	P1(mm)
StridedFormer [22]	27	4.01M	163	46.9
P-STMO [39]	27	4.6M	164	46.1
MHFormer [23]	27	18.92M	1000	45.9
MixSTE [52]	27	33.61M	15402	45.1
STCFormer	27	4.75M	2173	44.1
StridedFormer [22]	81	4.06M	392	45.4
P-STMO [39]	81	5.4M	493	44.1
MHFormer [23]	81	19.67M	1561	44.5
MixSTE [52]	81	33.61M	46208	42.7
STCFormer	81	4.75M	6520	42.0
StridedFormer [22]	243	4.23M	1372	44.0
P-STMO [39]	243	6.7M	1737	42.8
MHFormer [23]	243	24.72M	4812	43.2
MixSTE [52]	243	33.61M	138623	40.9
STCFormer	243	4.75M	19561	41.0
STCFormer-L	243	18.91M	78107	40.5

as input. **Spatial Attention** and **Temporal Attention** solely exploit the spatial pathway and temporal pathway, respectively. **STC only** contains both pathways but without the positional embedding. **SPE₁**, **SPE₂** and **SPE** represent the two SPE positional embeddings and their combination, respectively. Table 5 details the contribution of each component towards the overall performance. STC only by considering both spatial and temporal correlations leads to the error drop over solely utilizing spatial attention and temporal attention by 218.5mm and 10.6mm, respectively. The result indicates the importance of modeling the correlations along two axes in parallel. The three positional embedding strategies, i.e., **SPE₁**, **SPE₂** and **SPE**, further contribute 0.6mm, 12.1mm and 12.9mm of error drop, respectively, proving the advances of involving the structure information.

In addition to the proposed **SPE₁**, we explore three other positional embedding functions, i.e., Absolute Positional Embedding (APE), Centrality Positional Embedding

Table 5. Performance contribution of each component in the proposed STCFormer on Human3.6M dataset.

		STC	SPE ₁	SPE ₂	P1 (mm)
Spatial Attention	#1				275.5
Temporal Attention	#2				67.6
STC only	#3	✓			57.0
+SPE ₁	#4	✓	✓		56.4
+SPE ₂	#5	✓		✓	44.9
+SPE	#6	✓	✓	✓	44.1

(CPE), and Symmetric Positional Embedding (SyPE). We refer the readers to read the supplementary materials for more details. In Table 6, we assess the performance impact of different positional embedding functions. In this experiment, the models take the estimated 2D poses by CPN of 9 frames as input. And the comparisons empirically show the superiority of the used SPE₁ (48.3mm vs. 48.7mm, 49.9mm, and 49.2mm).

Table 6. The P1 error comparisons with different positional embedding functions on Human3.6M dataset. The “Baseline” denotes the STCFormer without SPE₁. The best result in each column is marked in **red**.

		P1 (mm)
Baseline	#1	48.7
+SPE ₁	#2	48.3
+APE	#3	48.9
+CPE	#4	49.6
+SyPE	#5	49.2

4.6. Qualitative Analysis

In this section, we validate our STCFormer through attention visualization and 3D human pose estimation visualization. The examples are randomly selected from the evaluation set of Human3.6M.

Attention visualization. We visualize the spatial attention map and temporal attention map from the last STC block of STCFormer in Figure 5. As expected, the spatial attention map (Figure 5(a)) shows that our model learns different patterns between joints from the videos of different actions. Moreover, the temporal attention map in Figure 5(b) illustrates strong correlation across adjacent frames owing to the continuity of human actions.

Result visualization on Human3.6M. Figure 6 shows cases 3D human pose estimation results by STCFormer and the recent transformer-based approaches including StridedFormer [22], MHFormer [23] and P-STMO [39]. The three examples are randomly selected from the walking, posing and sitting actions in Human3.6M dataset. For each method, we draw the estimated 3D human pose and the ground-truth 3D coordinates in one figure, and calculate the average error. Overall, our STCFormer shows better reconstruction results across all three samples than the other three methods. Particularly, for the challenging action with complicated pose articulation like “sitting” (the third row), STC-

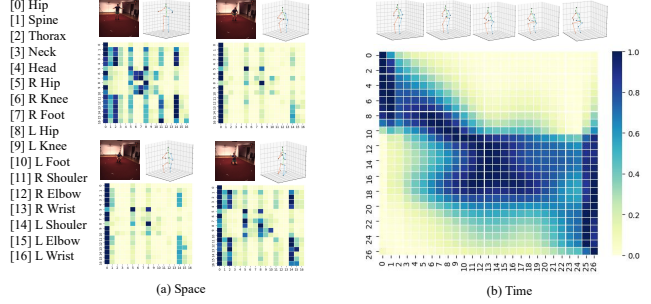


Figure 5. Visualizations of attention maps from the spatial and temporal attention modules in STCFormer. The x-axis and y-axis correspond to the queries and the predicted outputs, respectively.

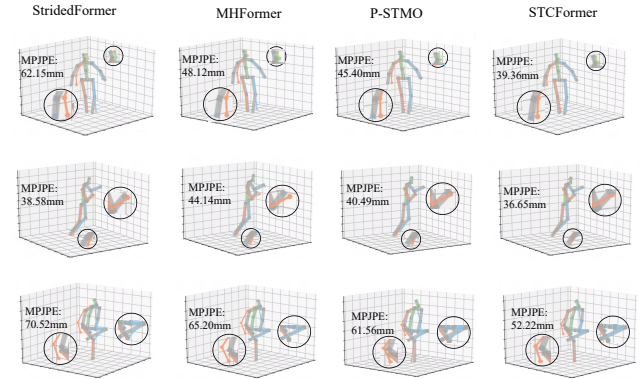


Figure 6. Examples of 3D pose estimation by StridedFormer [22], MHFormer [23], P-STMO [39] and our STCFormer. The gray skeleton is the ground-truth 3D pose. Blue, orange and green skeletons represent the left part, right part and torso of the estimated human body, respectively.

Former still estimates the 3D coordinates accurately and reconstructs the structurally plausible 3D pose.

5. Conclusion

We have presented Spatio-Temporal Criss-cross Transformer (STCFormer), which explores spatial correlation and temporal correlation in a two-pathway manner for 3D human pose estimation in videos. Particularly, STCFormer is built by stacking several STC blocks, each of which separates the joint features into two groups along the channel dimension, and models the spatial and temporal interactions on each group, respectively. By doing so, the receptive field of STC block is like a criss cross of spatial and temporal axes. Moreover, the STCFormer exploits the dynamic chain structure of human body to model local context, resulting in a new positional embedding function. The experiments conducted on two benchmarks demonstrate the effectiveness of STCFormer and good generalization ability compared to the state-of-the-art techniques.

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